

1995-2021 UNITED STATES HONEY PRODUCTION, SUPPLY AND DEMAND ANALYSIS

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Tableau Public Link:

https://public.tableau.com/app/profile/arnold.atchoe/viz/ArnoldAtchoe_HoneyProduction/1995-2021U_SHoneyProduction?publish=yes

GitHub Link: <https://github.com/kwameape123/U.S.-Honey-Production>

1. Section 1. Introduction

According to Wikipedia, the United States is among the top 10 countries in the world when it comes to honey production. Honey is a natural product made by bees. Bees visit flowers to collect nectar and bring it back to their hive to produce honey. Honeybees use honey as their primary source of energy and their instinct is to make more than their colony needs. Beekeepers harvest the excess and bottle it for consumption. The goal of this project is to analyze honey production, supply and demand in the United States from 1995 to 2021. This study specifically summarizes national Key Performance Indicators (KPIs) for honey production and observes trends of independent variables (KPIs) and how they influence the trend of dependent variables (KPIs) over time at the national level and for the top 10 producing states. The project also includes a dashboard to allow users to analyze honey production and related trends at the regional and state level. Associate visualizations, dashboards and stories can be access at https://public.tableau.com/app/profile/arnold.atchoe/viz/ArnoldAtchoe_HoneyProduction/1995-2021U_SHoneyProduction?publish=yes. Source files used in this project can be accessed at <https://github.com/kwameape123/U.S.-Honey-Production>.

2. Section 2. Data Description

US Honey Production 1995-2021 is an agricultural dataset, and it was obtained from Kaggle at <https://www.kaggle.com/datasets/mohitpoudel/us-honey-production-19952021/data> as a Comma Separated Values (CSV) file. This Kaggle dataset was curated using yearly honey production data reported by the National Agricultural Statistics Service of the United States Department of Agriculture. The original dataset in its raw form obtained from Kaggle has 1116 rows (1 header row and 1115 observations) and 9 columns. Description of original dataset columns are as follows:

- **Index number:** Unique Identification number for each observation in the dataset.
- **State:** Name of State.
- **Colonies number:** Number of honey producing colonies. Honey producing colonies are the maximum number of colonies in the year. It is possible to harvest honey from colonies that did not survive the entire year.
- **Yield per colony:** Units of honey produced per colony in pounds.
- **Production:** Total Production is the number of colonies multiplied by yield per colony.
- **Stocks:** Units of honey held in stock at the end of each year in pounds.
- **Average price:** Refers to average price per pound based on expanded sales. Unit is dollars.

- **Value of production:** Value of production is obtained by multiplying production and average_price. Unit is in dollars.
- **Year:** Year of which data is tabulated.

3. Section 3. Data Cleaning Strategies

- The U.S honey production dataset for 1995 to 2021 in its original form from Kaggle had some calculation errors and inconsistencies as follows:
 - Production column which is a product of colonies_numbers and yield_per_colony contained wrong values.
 - Average price was recorded in cents for 1995 to 2017 and in dollars for 2018 to 2021.
 - Because value_of_production depends on average prices, the values for value_of_production were incorrect.
- To correct the problems listed above, the following measures were taken with Microsoft Excel:
 - The original production column was deleted and a new production column generated by multiplying colonies_numbers by yield_per_colony. Results were copied and pasted as values to get rid of formulas in cells.
 - From 1995 to 2017 prices were converted to dollars by dividing value in cents by 100 while 2018 to 2021 values remained the same. Results were copied and pasted as values to get rid of formulas in cells.
 - The original value_of_production column was deleted, and a new value_of_production column was generated by multiplying production by average_prices. Results were copied and pasted as values to get rid of formulas in cells.
 - As part of the data cleaning exercise, a value_of_stocks column was generated by multiplying the stocks column by the average_price column.
 - As part of the data cleaning process. A revenue column was generated by finding the difference between value_of_production and value_of_stocks.

4. Section 4. Clean Dataset

Performing data cleaning on the original dataset from Kaggle resulted in a dataset with the following 11 columns and same number of rows:

- **Index number:** Unique Identification number for each observation in the dataset.
- **State:** Name of State.
- **Colonies_number:** Number of honey producing colonies. Honey producing colonies are the maximum number of colonies in the year. It is possible to harvest honey from colonies that did not survive the entire year.
- **Yield per colony:** Units of honey produced per colony in pounds.
- **Production:** Total Production is number of colonies multiplied by yield per colony.
- **Stocks:** Units of honey held in stock at the end of each year in pounds.
- **Average price:** Refers to average price per pound based on expanded sales. Unit is dollars.
- **Value of production:** Value of production is obtained by multiplying production and average_price. Unit is dollars.

- **Year:** Year of which data is tabulated.
- **Value of stocks:** This column represents the product of average_price and stocks.
- **Revenue:** This column represents the difference between value_of_production and value_of_stocks.

To facilitate analysis and visualization for this project, columns, summary tables and values were generated in Tableau. These columns, summary tables and values include:

- **Regions:** Represent the administrative regions of the National Agricultural Statistics Service of the United States Department of Agriculture.
- **Average State Production:** This value represents the average quantity of honey in pounds produced state-wise.
- **Colonies number 1995-2021:** This summary table presents the number of colonies for each year from 1995 to 2021. It was useful in calculating average number of colonies nationally.
- **Demand:** This summary table presents the units of honey in pounds sold for each year from 1995 to 2021. It was useful in visualizing the trend of demand.
- **State Production Sum:** This summary table presents the units of honey produced in pounds for each state for 1995 to 2021. It was useful in determining or calculating average state production.
- **Time Production Sum:** This summary table presents the units of honey produced in pounds for each year from 1995 to 2021. It was useful in establishing the relationship between demand and production.
- **Yield per colony 1995-2021:** This summary table presents the average yield per colony for each year from 1995 to 2021. It was useful in observing average yield per colony over the years from 1995 to 2021.

index	state	colonies_n	yield_per_c	production	stocks	average_p	value_of_p	value_of_s	revenue	year
0	Alabama	16000	58	928000	28000	0.62	575360	17360	558000	1995
1	Arizona	52000	79	4108000	986000	0.68	2793440	670480	2122960	1995
2	Arkansas	50000	60	3000000	900000	0.64	1920000	576000	1344000	1995
3	California	420000	93	39060000	4687000	0.6	23436000	2812200	20623800	1995
4	Colorado	45000	60	2700000	1404000	0.68	1836000	954720	881280	1995
5	Florida	230000	86	19780000	1780000	0.63	12461400	1121400	11340000	1995
6	Georgia	70000	62	4340000	260000	0.69	2994600	179400	2815200	1995
7	Hawaii	8000	129	1032000	103000	0.55	567600	56650	510950	1995
8	Idaho	125000	48	6000000	1020000	0.65	3900000	663000	3237000	1995
9	Illinois	11000	74	814000	212000	1.02	830280	216240	614040	1995
10	Indiana	12000	63	756000	166000	0.68	514080	112880	401200	1995

First 10 rows of the clean dataset.

5. Section 5. Visualization Tool

Tableau was used to create various visualizations, a dashboard, and a data story. It was selected as the visualization tool because it provides an easy-to-use interface and intuitive menu options that make it simple to build visualizations, dashboards, and data stories.

6. Section 6. Visualization and Stories

➤ Section 6.1 United States Honey Production Key Performance Indicators

Goal: Determine what the average number of colonies, the average yield per colony, the units (in pounds) and value (in dollars) of honey produced, units (in pounds) and value (in dollars) of honey in stock at the end of the year and revenue for the United States over the period 1995 to 2021.

Approach: A summary table was selected as the visualization of choice, even though it is not a traditional chart. This format allows the audience to quickly identify and interpret the key indicators of the U.S. honey market from 1995 to 2021.

Chart and Story:

United States Honey Production Statistics.	
Avg. Colonies_ number_1995-2021	2,578,481
Avg. Yield_per_ colony_1995-2021	60
Production	4,573,158,000
Revenue	4,620,919,850
Stocks	1,307,477,000
Value Of Production	6,319,173,080
Value Of Stocks	1,698,253,230

Caption

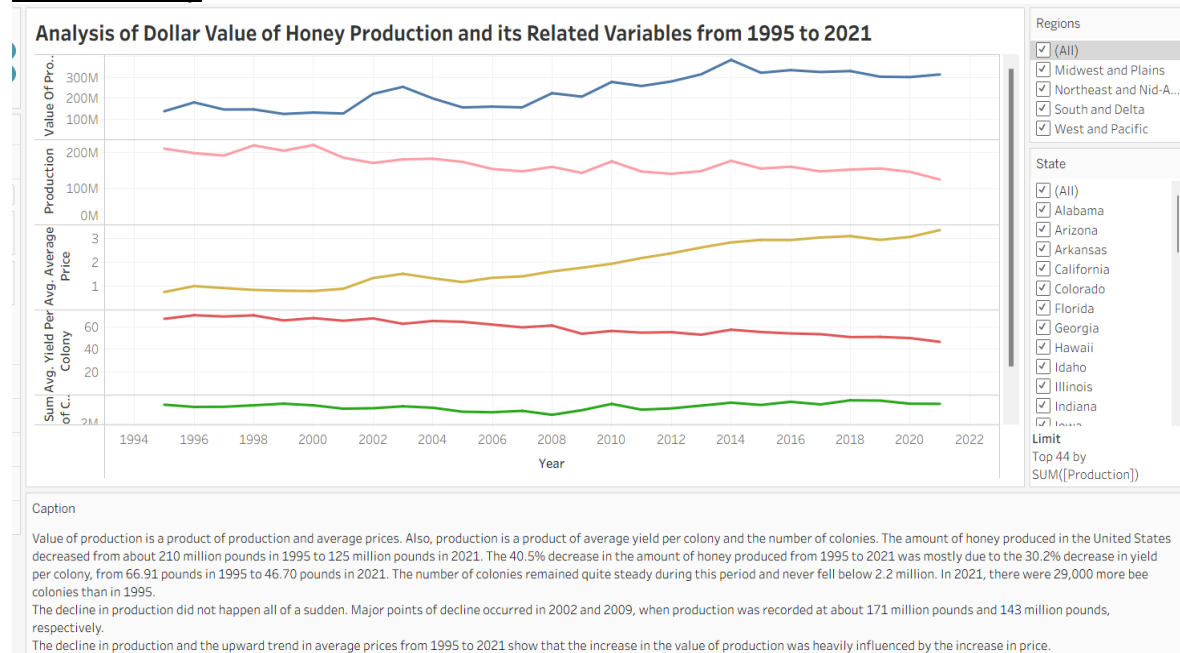
Data collected by the National Agricultural Statistics Service (NASS) of the USDA from 1995 to 2021 indicates that the United States produced approximately 4.6 billion pounds of honey, worth about \$6.3 billion. To achieve this level of production over this period, the United States had to maintain, on average, about 2.6 million bee colonies each year, with an average yield of 60 pounds per colony. Cumulatively, approximately 1.3 billion pounds out of the 4.6 billion pounds of honey produced was in stock at the end of the year from 1995 to 2021. This inventory was worth about \$1.7 billion. This excess showed that supply exceeded demand, resulting in approximately \$4.6 billion in honey value that was not held in year-end inventory.

➤ Section 6.2.1 Analysis of Dollar Value of Honey Production and its Related Variables from 1995 to 2021.

Goal: To determine how number of bee colonies, average yield per colony and average prices affect production and the value of production for the United States over the period 1995 to 2021.

Approach: Multiple panel line charts were presented side by side to visualize and analyze the changes in independent variables namely number of colonies, average yield per colony and average prices to determine how they affected the changes in dependent variables production and value of production.

Chart and Story:

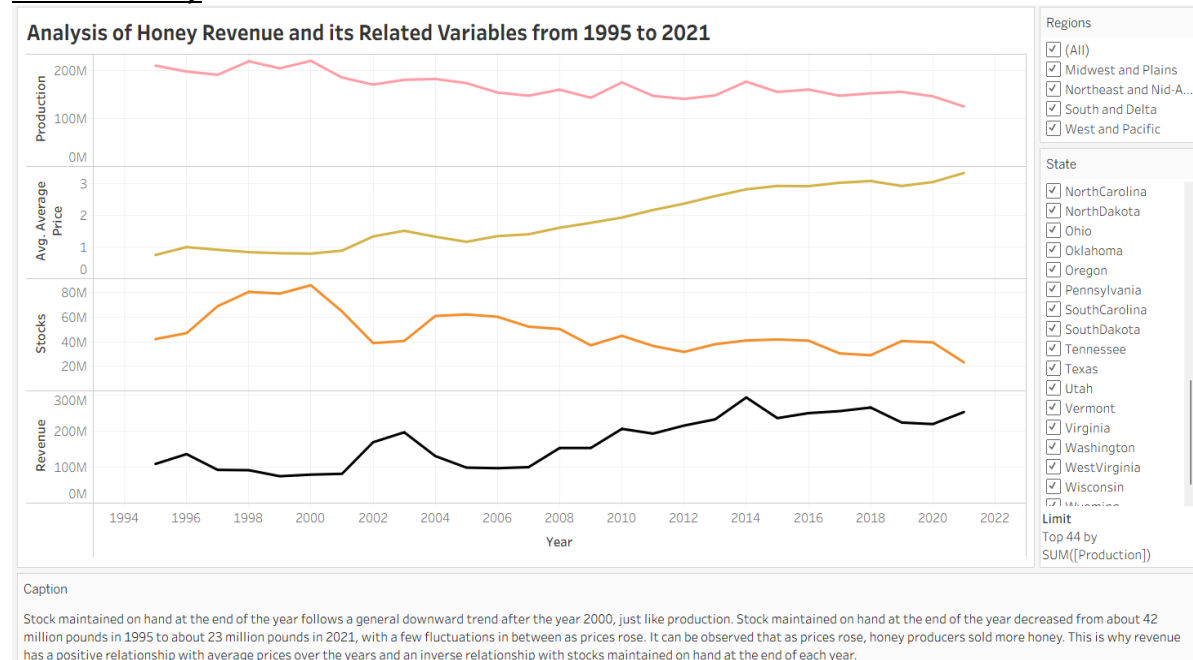


➤ Section 6.2.2 Analysis of Honey Revenue and its Related Variables from 1995 to 2021.

Goal: To determine how stocks withheld, production and average prices affect revenue for the United States over the period 1995 to 2021.

Approach: Multiple panel line charts were presented side by side to visualize and analyze the changes in variables stocks on hand, production and average prices to determine how they affected the changes in revenue.

Chart and Story:

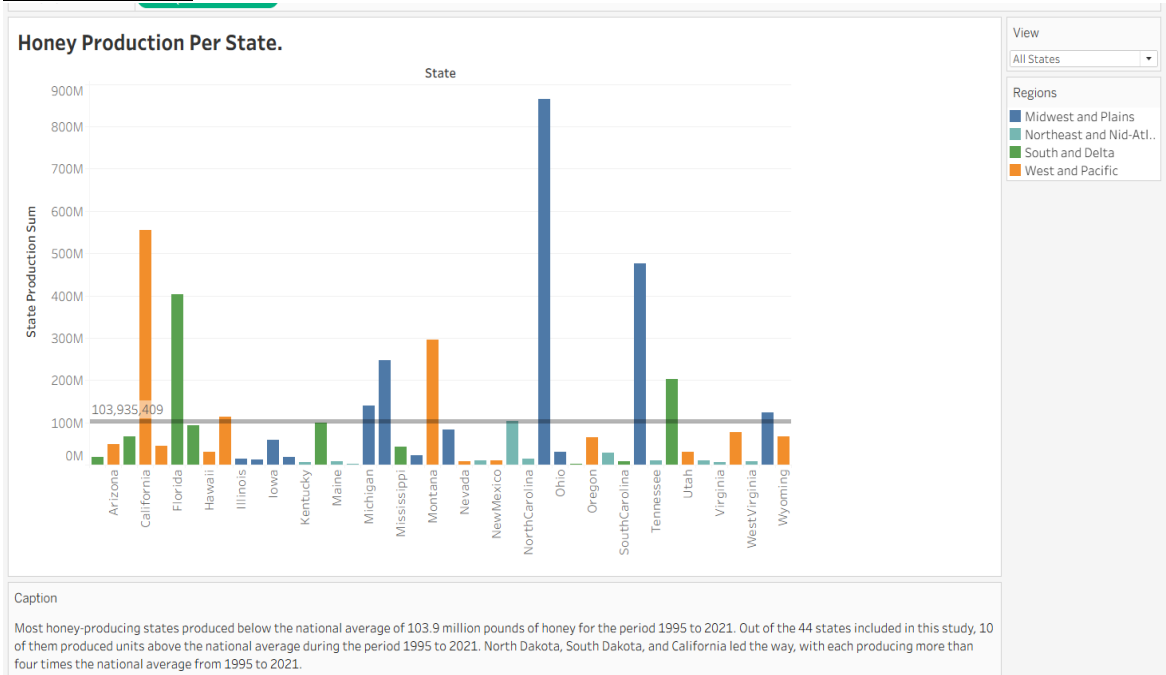


➤ Section 6.3 Honey Production Per State.

Goal: To determine how many units of honey each honey producing state produced and how their production level compares to the average production level for 1995 to 2021.

Approach: A column chart was generated to show production levels for each state for 1995-2021. A reference line representing average production levels for 1995 to 2021 was incorporated into the column chart to drive a benchmark chart.

Chart and Story:



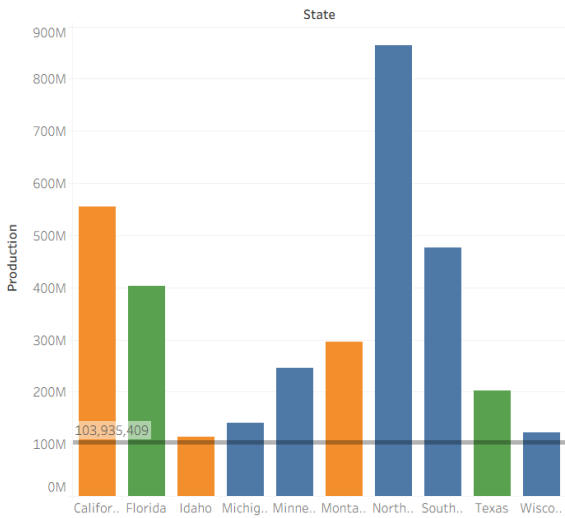
➤ **Section 6.4 Top 10 Honey Producing States**

Goal: To determine how many units of honey each of the top 10 honey producing state produced and how their production level compares to the average production level for 1995 to 2021.

Approach: A column chart was generated to show production levels for each of the top 10 states for 1995-2021. A reference line representing average production levels for 1995 to 2021 was incorporated into the column chart to drive a benchmark chart.

Chart and Story:

Top 10 Honey Producing States



Regions

☒ (All)

☒ Midwest and Plains

☒ Northeast and Mid-A...

☒ South and Delta

☒ West and Pacific

Regions

☒ Midwest and Plains

☒ South and Delta

☒ West and Pacific

caption

California, Florida, Idaho, Michigan, Minnesota, Montana, North Dakota, South Dakota, Texas, and Wisconsin were the top honey producers in the United States considering the period from 1995 to 2021. Fifty percent of these top 10 producers are in the Midwest and Plains Region, and together they accounted for about 47% of the units produced by the top producers from 1995 to 2021.

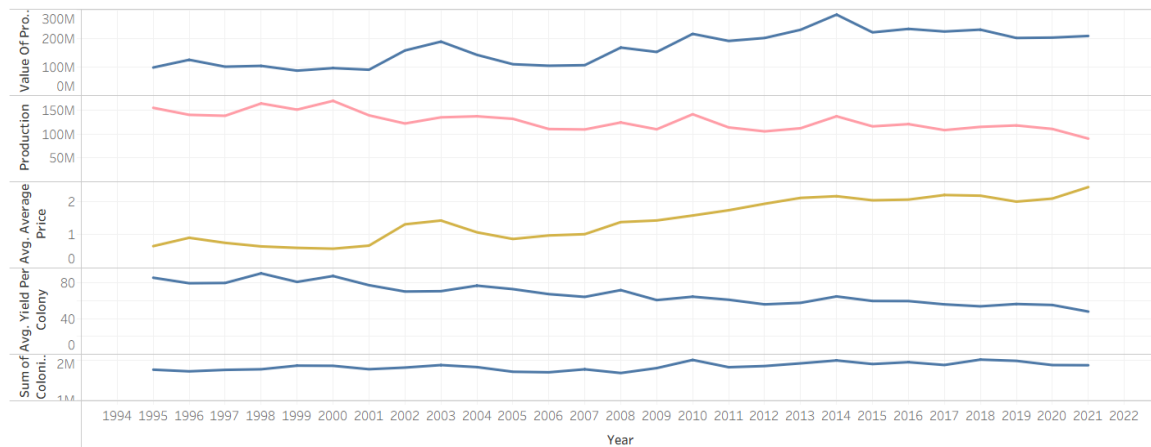
➤ **Section 6.5 Analysis of Honey's Value of Production and its Related Variables of Top 10 Producing States from 1995 to 2021.**

Goal: To determine how number of bee colonies, average yield per colony and average prices affect production and the value of production for the top 10 honey producers over the period 1995 to 2021.

Approach: Multiple panel line charts were presented side by side to visualize and analyze the changes in independent variables namely number of colonies, average yield per colony and average prices to determine how they affected the changes in dependent variables production and value of production.

Chart and Story:

Analysis of Honey's Value of Production and its Related Variables of Top 10 Producing States from 1995 to 2021



Caption

The production trend for the top 10 honey producers over the period 1995 to 2021 mirrors the national production trend. There was a general decline in units produced from 1995 to 2021 of about 41.6%. This was due to a 44.2% decline in yield per colony over this period, even though the number of colonies maintained trended upward. The major decline in production and yield per colony began in the year 2000.

The upward trend in average prices over the period from 1995 to 2021 was the major influence on the growth in the value of production.

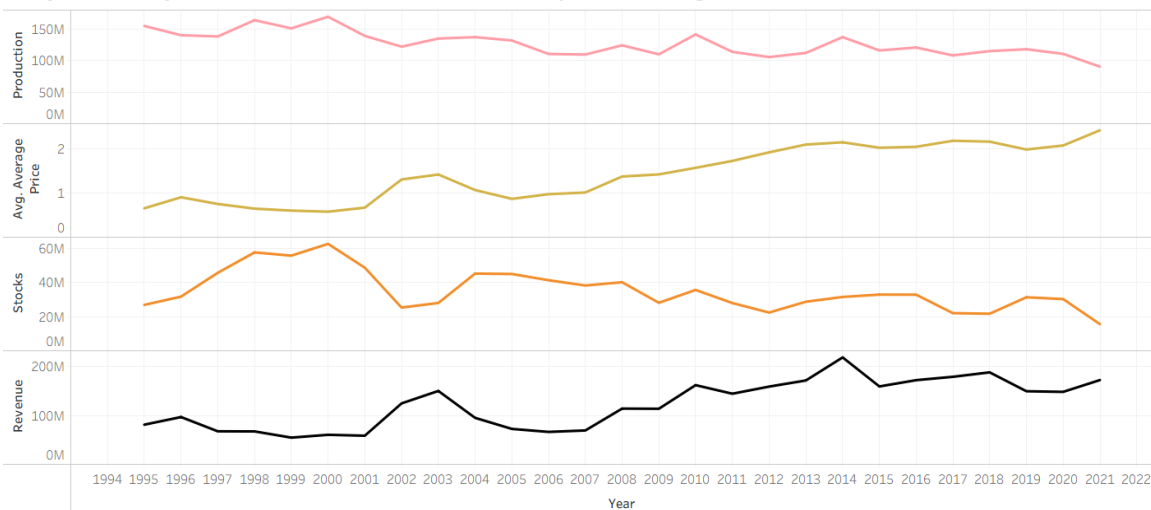
➤ **Section 6.6 Analysis of Honey Revenue and its Related Variables of Top 10 Producing States from 1995 to 2021.**

Goal: To determine how stocks withheld, production and average prices affect revenue for the top 10 honey producing states over the period 1995 to 2021.

Approach: Multiple panel line charts were presented side by side to visualize and analyze the changes in variables stocks on hand, production and average prices to determine how they affected the changes in revenue.

Chart and Story:

Analysis of Honey Revenue and its Related Variables of Top 10 Producing States from 1995 to 2021



Caption

Compared to the national level, stock maintained on hand at the end of the year by the top 10 honey producers during the study period also follows a general downward trend, just like production, after the year 2000. Revenue and average prices follow a downward trend and an upward trend prior to 2001 and beginning in 2001, respectively, contrary to the downward trend in production and stocks withheld at the end of the year. This indicates that as prices increase, revenue also increases. Therefore, prices are the major driver of the increase in revenue.

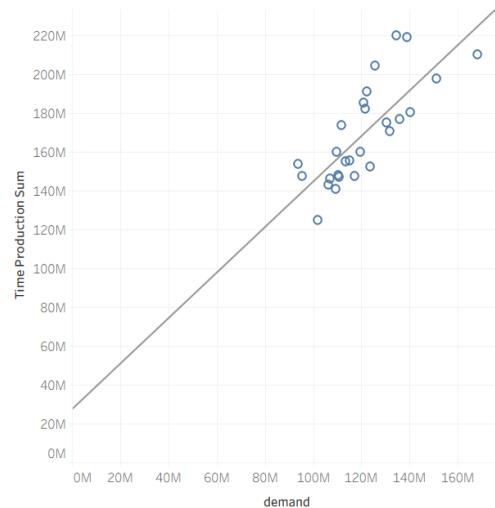
➤ **Section 6.7 Relationship between Demand and Units Produced for Top 10 Honey Producers.**

Goal: To understand the relationship between demand and production for the top 10 honey producing states.

Approach: First a scatter plot was developed to visualize the relationship between production and demand of the top 10 honey producers. Subsequently a trend was added to show the associated mathematical relationship.

Chart and Story:

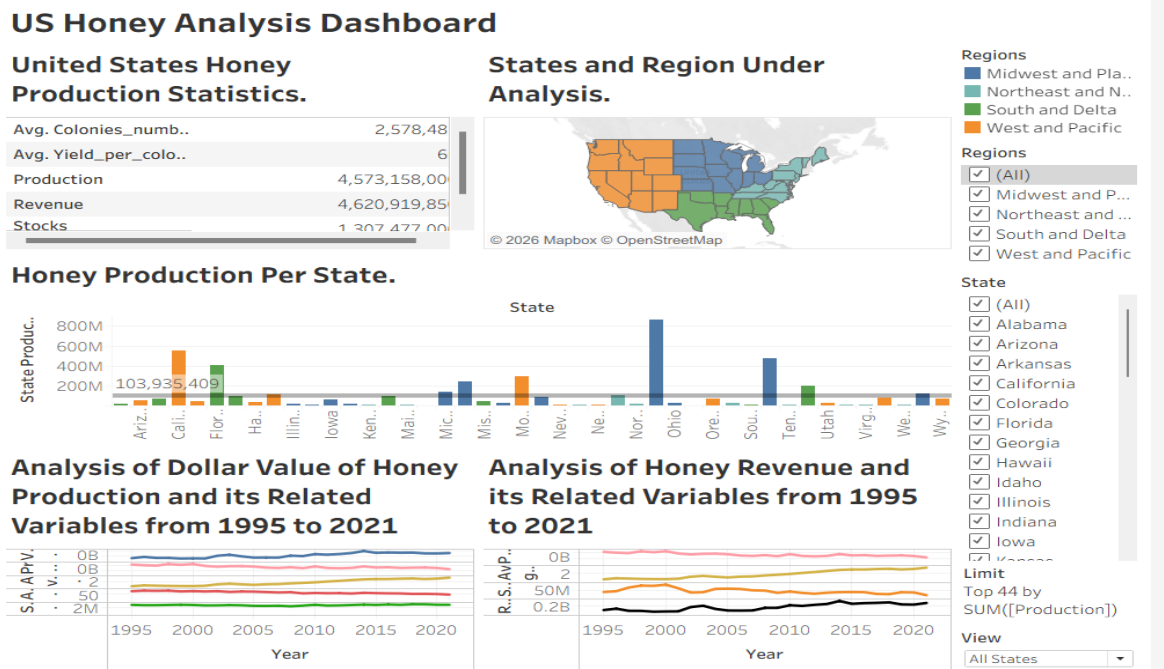
Relationship between Demand and Units Produced For Top 10 Honey Producers.



Caption

For the top 10 honey producers from 1995 to 2021, production tends to increase or decrease with demand. Demand accounts for 59.6% of the changes in production.

7. Section 7. Conclusions



The analysis of honey production in the United States from 1995 to 2021 shows that the honey industry experienced a general decline in production, while the value of production increased over the same period. Production declined mainly due to the reduction in average yield per colony, while the number of bee colonies remained relatively stable and, in some cases, increased. Stocks maintained at the end of the year also followed a downward trend, especially after 2000. At the same time, average honey prices generally trended upward, making price increases a major factor influencing the growth in the value of production and revenue.

The top 10 honey-producing states followed a production trend like the national trend, with production declining mainly due to the reduction in average yield per colony despite the number of colonies trending upward. States in the Midwest-Plains region made up about half of the top 10 producers and remained an important region for the honey industry because they consistently produced significantly more honey than other regions. The relationship between production and demand also shows that demand had an important influence on production, accounting for approximately 59.6% of the changes in production among the top 10 honey-producing states.

Overall, the U.S. honey industry became more dependent on higher prices to maintain and increase the value of production as physical production declined. The stability or increase in the number of colonies was not enough to prevent the decline in production because yield per colony decreased substantially. Therefore, improving productivity per colony, responding to changes in demand, and maintaining favorable prices are important for the growth and sustainability of the U.S. honey industry. Further analysis of 2022 to 2026 will be beneficial in determining whether the decline in production

continues due to declining average yield per colony and could lead to an investigation into whether environmental or other factors are contributing to the decline in yield.